Isolated Black Hole Horizons

Quasi-Local Understanding of Horizon Physics

Nishkal Rao

Indian Institute of Science Education and Research, Pune
nishkal.rao@students.iiserpune.ac.in

Supervisor:	Dr. Badri Krishnan, Radboud University/Albert Einstein Institute.
Co-supervisor:	Dr. Sukanta Bose, Washington State University.
	Dr. Apratim Ganguly, The Inter-University Centre for Astronomy and Astrophysics, Pune.
Expert:	Dr. Suneeta Vardarajan, Indian Institute of Science Education and Research, Pune.

Overview

1. Preliminaries
2. Isolated Horizons
3. Probing Near-Horizon Structure
4. Tidal Deformabilities

Outline

1. Preliminaries
2. Isolated Horizons
3. Probing Near-Horizon Structure
4. Tidal Deformabilities

Newman-Penrose Tetrad

Orthonormal conditions,

$$\boldsymbol{\ell}\cdot\boldsymbol{m}=\boldsymbol{\ell}\cdot\bar{\boldsymbol{m}}=\boldsymbol{n}\cdot\boldsymbol{m}=\boldsymbol{n}\cdot\bar{\boldsymbol{m}}=0. \tag{1}$$

Null requirements,

$$\boldsymbol{\ell}\cdot\boldsymbol{\ell}=\boldsymbol{m}\cdot\boldsymbol{m}=\bar{\boldsymbol{m}}\cdot\bar{\boldsymbol{m}}=\boldsymbol{n}\cdot\boldsymbol{n}=0. \tag{2}$$

Normalization conditions,

$$\boldsymbol{\ell}\cdot\boldsymbol{n}=-1,\quad \boldsymbol{m}\cdot\bar{\boldsymbol{m}}=1. \tag{3}$$

Spacetime metric,

$$\boldsymbol{g}=-\boldsymbol{\ell}\otimes\boldsymbol{n}-\boldsymbol{n}\otimes\boldsymbol{\ell}+\boldsymbol{m}\otimes\bar{\boldsymbol{m}}+\bar{\boldsymbol{m}}\otimes\boldsymbol{m}, \tag{4}$$

Spin Coefficients

Directional Derivatives: $D = \ell^\mu \nabla_\mu$, $\Delta = n^\mu \nabla_\mu$, $\delta = m^\mu \nabla_\mu$.

$$D\boldsymbol{\ell} = (\varepsilon + \bar{\varepsilon})\boldsymbol{\ell} - \bar{\kappa}\boldsymbol{m} - \kappa\bar{\boldsymbol{m}}, \tag{5a}$$

$$D\boldsymbol{n} = -(\epsilon + \bar{\epsilon})\boldsymbol{n} + \pi\boldsymbol{m} + \bar{\pi}\bar{\boldsymbol{m}}, \tag{5b}$$

$$D\boldsymbol{m} = \bar{\pi}\boldsymbol{\ell} - \kappa\boldsymbol{n} + (\epsilon - \bar{\epsilon})\boldsymbol{m}, \tag{5c}$$

$$\Delta\boldsymbol{\ell} = (\gamma + \bar{\gamma})\boldsymbol{\ell} - \bar{\tau}\boldsymbol{m} - \tau\bar{\boldsymbol{m}}, \tag{5d}$$

$$\Delta\boldsymbol{n} = -(\gamma + \bar{\gamma})\boldsymbol{n} + \nu\boldsymbol{m} + \bar{\nu}\bar{\boldsymbol{m}}, \tag{5e}$$

$$\Delta\boldsymbol{m} = \bar{\nu}\boldsymbol{\ell} - \tau\boldsymbol{n} + (\gamma - \bar{\gamma})\boldsymbol{m}, \tag{5f}$$

$$\delta\boldsymbol{\ell} = (\bar{\alpha} + \beta)\boldsymbol{\ell} - \bar{\varrho}\boldsymbol{m} - \sigma\bar{\boldsymbol{m}} \tag{5g}$$

$$\delta\boldsymbol{n} = -(\bar{\alpha} + \beta)\boldsymbol{n} + \mu\boldsymbol{m} + \bar{\lambda}\bar{\boldsymbol{m}}, \tag{5h}$$

$$\delta\boldsymbol{m} = \bar{\lambda}\boldsymbol{\ell} - \sigma\boldsymbol{n} + (\beta - \bar{\alpha})\boldsymbol{m}, \tag{5i}$$

$$\bar{\delta}\boldsymbol{m} = \bar{\mu}\boldsymbol{\ell} - \varrho\boldsymbol{n} + (\alpha - \bar{\beta})\boldsymbol{m}. \tag{5j}$$

Exterior Derivatives

$$
\begin{aligned}
d\boldsymbol{\ell} &= (\epsilon + \bar{\epsilon})\boldsymbol{\ell} \wedge \boldsymbol{n} + [\bar{\tau} - (\alpha + \bar{\beta})]\boldsymbol{\ell} \wedge \boldsymbol{m} + [\tau - (\bar{\alpha} + \beta)]\boldsymbol{\ell} \wedge \bar{\boldsymbol{m}} \\
&\quad + \bar{\kappa}\boldsymbol{n} \wedge \boldsymbol{m} + \kappa\boldsymbol{n} \wedge \bar{\boldsymbol{m}} + (\bar{\varrho} - \varrho)\boldsymbol{m} \wedge \bar{\boldsymbol{m}}, && \text{(6a)} \\
d\boldsymbol{n} &= (\gamma + \bar{\gamma})\boldsymbol{\ell} \wedge \boldsymbol{n} - \nu\boldsymbol{\ell} \wedge \boldsymbol{m} - \bar{\nu}\boldsymbol{\ell} \wedge \bar{\boldsymbol{m}} - [\pi - (\alpha + \bar{\beta})]\boldsymbol{n} \wedge \boldsymbol{m} \\
&\quad - [\bar{\pi} - (\bar{\alpha} + \beta)]\boldsymbol{n} \wedge \bar{\boldsymbol{m}} + (\bar{\mu} - \mu)\boldsymbol{m} \wedge \bar{\boldsymbol{m}}, && \text{(6b)} \\
d\boldsymbol{m} &= (\tau + \bar{\pi})\boldsymbol{\ell} \wedge \boldsymbol{n} - [\bar{\mu} + (\gamma - \bar{\gamma})]\boldsymbol{\ell} \wedge \boldsymbol{m} + \bar{\lambda}\boldsymbol{\ell} \wedge \bar{\boldsymbol{m}} \\
&\quad + [\varrho - (\epsilon - \bar{\epsilon})]\boldsymbol{n} \wedge \boldsymbol{m} + \sigma\boldsymbol{n} \wedge \bar{\boldsymbol{m}} + (\bar{\alpha} - \beta)\boldsymbol{m} \wedge \bar{\boldsymbol{m}}. && \text{(6c)}
\end{aligned}
$$

Weyl Tensor

$$\Psi_0 = C_{\mu\nu\alpha\beta}\ell^\mu m^\nu \ell^\alpha m^\beta, \tag{7a}$$

$$\Psi_1 = C_{\mu\nu\alpha\beta}\ell^i n^\nu \ell^\alpha m^\beta, \tag{7b}$$

$$\Psi_2 = C_{\mu\nu\alpha\beta}\ell^\mu m^\nu \bar{m}^\alpha n^\beta, \tag{7c}$$

$$\Psi_3 = C_{\mu\nu\alpha\beta}\ell^i n^\nu \bar{m}^\alpha n^\beta, \tag{7d}$$

$$\Psi_4 = C_{\mu\nu\alpha\beta} n^\mu \bar{m}^\nu n^\alpha \bar{m}^\beta. \tag{7e}$$

Curvature Projections

$$\Lambda = \frac{R}{24}, \tag{8a}$$

$$\Phi_{00} = \frac{1}{2} R_{\mu\nu} \ell^\mu \ell^\nu, \tag{8b}$$

$$\Phi_{11} = \frac{1}{4} R_{\mu\nu} (\ell^\mu n^\nu + m^\mu \bar{m}^\nu), \tag{8c}$$

$$\Phi_{22} = \frac{1}{2} R_{\mu\nu} n^\mu n^\nu, \tag{8d}$$

$$\Phi_{01} = \frac{1}{2} R_{\mu\nu} \ell^\mu m^\nu, \tag{8e}$$

$$\Phi_{10} = \bar{\Phi}_{01} = \frac{1}{2} R_{\mu\nu} \ell^\mu \bar{m}^\nu, \tag{8f}$$

$$\Phi_{02} = \frac{1}{2} R_{\mu\nu} m^\mu m^\nu, \tag{8g}$$

$$\Phi_{20} = \bar{\Phi}_{02} = \frac{1}{2} R_{\mu\nu} \bar{m}^\mu \bar{m}^\nu, \tag{8h}$$

$$\Phi_{12} = \frac{1}{2} R_{\mu\nu} n^\mu m^\nu, \tag{8i}$$

$$\Phi_{21} = \frac{1}{2} R_{\mu\nu} n^\mu \bar{m}^\nu. \tag{8j}$$

Radial Field Equations

$$\Delta\pi - D\nu = -\mu\left(\pi + \bar{\tau}\right) - \lambda\left(\bar{\pi} + \tau\right) - \pi\left(\gamma - \bar{\gamma}\right) + \nu\left(3\epsilon + \bar{\epsilon}\right) - \Psi_3 - \Phi_{21}, \tag{9a}$$

$$\Delta\beta - \delta\gamma = \gamma\left(\bar{\alpha} + \beta - \tau\right) - \mu\tau + \sigma\nu + \epsilon\bar{\nu} + \beta\left(\gamma - \bar{\gamma} - \mu\right) - \alpha\bar{\lambda} - \Phi_{12}, \tag{9b}$$

$$\Delta\alpha - \bar{\delta}\gamma = \nu\left(\varrho + \epsilon\right) - \lambda\left(\tau + \beta\right) + \alpha\left(\bar{\gamma} - \bar{\mu}\right) + \gamma\left(\bar{\beta} - \bar{\tau}\right) - \Psi_3, \tag{9c}$$

$$\Delta\lambda - \bar{\delta}\nu = \lambda\left(\bar{\gamma} - 3\gamma - \mu - \bar{\mu}\right) + \nu\left(3\alpha + \bar{\beta} + \pi - \bar{\tau}\right) - \Psi_4, \tag{9d}$$

$$\Delta\mu - \delta\nu = -\mu^2 - |\lambda|^2 - \mu\left(\gamma + \bar{\gamma}\right) + \bar{\nu}\pi + \nu\left(\bar{\alpha} + 3\beta - \tau\right) - \Phi_{22}, \tag{9e}$$

$$\Delta\varrho - \bar{\delta}\tau = -\varrho\bar{\mu} - \sigma\lambda + \tau\left(\bar{\beta} - \alpha - \bar{\tau}\right) + \varrho\left(\gamma + \bar{\gamma}\right) + \nu\kappa - \Psi_2 - 2\Lambda, \tag{9f}$$

$$\Delta\sigma - \delta\tau = -\mu\sigma - \bar{\lambda}\varrho + \tau\left(\tau + \beta - \bar{\alpha}\right) + \sigma\left(3\gamma - \bar{\gamma}\right) + \kappa\bar{\nu} - \Phi_{02}, \tag{9g}$$

$$\Delta\kappa - D\tau = -\varrho\left(\tau + \bar{\pi}\right) - \sigma\left(\bar{\tau} + \pi\right) - \tau\left(\epsilon - \bar{\epsilon}\right) + \kappa\left(3\gamma + \bar{\gamma}\right) - \Psi_1 - \Phi_{01}, \tag{9h}$$

$$\Delta\epsilon - D\gamma = -\alpha\left(\tau + \bar{\pi}\right) - \beta\left(\bar{\tau} + \pi\right) + 2\gamma\epsilon + \gamma\bar{\epsilon} + \epsilon\bar{\gamma} - \tau\pi + \nu\kappa - \Psi_2 - \Phi_{11} + \Lambda, \tag{9i}$$

Evolution Field Equations

$$D\varrho - \bar{\delta}\kappa = \varrho^2 + |\sigma|^2 + \varrho\left(\epsilon + \bar{\epsilon}\right) - \bar{\kappa}\tau + \kappa\left(\pi - 3\alpha - \bar{\beta}\right) + \Phi_{00}, \tag{10a}$$

$$D\sigma - \delta\kappa = \sigma\left(3\epsilon - \bar{\epsilon} + \varrho + \bar{\varrho}\right) - \kappa\left(\bar{\pi} + \tau + 3\bar{\beta} + \alpha\right) + \Psi_0, \tag{10b}$$

$$D\alpha - \bar{\delta}\epsilon = \alpha\left(\varrho + \bar{\epsilon} - 2\epsilon\right) + \beta\bar{\sigma} - \bar{\beta}\epsilon - \kappa\lambda - \bar{\kappa}\gamma + \pi\left(\epsilon + \varrho\right) + \Phi_{10}, \tag{10c}$$

$$D\beta - \delta\epsilon = \sigma\left(\alpha + \pi\right) + \beta\left(\bar{\varrho} - \bar{\epsilon}\right) - \kappa\left(\mu + \gamma\right) + \epsilon\left(\bar{\pi} - \bar{\alpha}\right) + \Psi_1, \tag{10d}$$

$$D\lambda - \bar{\delta}\pi = \varrho\lambda + \bar{\sigma}\mu + \pi\left(\pi + \alpha - \bar{\beta}\right) - \nu\bar{\kappa} + \lambda\left(\bar{\epsilon} - 3\epsilon\right) + \Phi_{20}, \tag{10e}$$

$$D\mu - \delta\pi = \bar{\varrho}\mu + \sigma\lambda + \pi\left(\bar{\pi} - \bar{\alpha} + \beta\right) - \mu\left(\epsilon + \bar{\epsilon}\right) - \nu\kappa + \Psi_2 + 2\Lambda, \tag{10f}$$

Angular Field Equations

$$\delta\alpha - \bar{\delta}\beta = \mu\varrho - \lambda\sigma + |\alpha|^2 + |\beta|^2 - 2\alpha\beta + \gamma\left(\varrho - \bar{\varrho}\right) + \epsilon\left(\mu - \bar{\mu}\right) - \Psi_2 + \Phi_{11} + \Lambda, \tag{11a}$$

$$\delta\varrho - \bar{\delta}\sigma = \varrho\left(\bar{\alpha} + \beta\right) + \sigma\left(\bar{\beta} - 3\alpha\right) + \tau\left(\varrho - \bar{\varrho}\right) + \kappa\left(\mu - \bar{\mu}\right) - \Psi_1 + \Phi_{01}, \tag{11b}$$

$$\delta\lambda - \bar{\delta}\mu = \nu\left(\varrho - \bar{\varrho}\right) + \pi\left(\mu - \bar{\mu}\right) + \mu\left(\alpha + \bar{\beta}\right) + \lambda\left(\bar{\alpha} - 3\beta\right) - \Psi_3 + \Phi_{21}. \tag{11c}$$

Bianchi Identities I

$$
\begin{aligned}
D\Psi_2 - \bar{\delta}\Psi_1 - \bar{\delta}\Phi_{01} + \Delta\Phi_{00} = & -\lambda\Psi_0 + 2\Psi_1(\pi - \alpha) + 3\varrho\Psi_2 - 2\kappa\Psi_3 - 2\Phi_{01}(\alpha + \bar{\tau}) \\
& + 2\varrho\Phi_{11} + \bar{\sigma}\Phi_{02} - \Phi_{00}(\bar{\mu} - 2\gamma - 2\bar{\gamma}) - 2\tau\Phi_{10} \\
& - 2D\Lambda, \qquad (12a) \\
\bar{\delta}\Psi_2 - D\Psi_3 - \delta\Phi_{20} + D\Phi_{21} = & 2\lambda\Psi_1 - 3\pi\Psi_2 + 2\Psi_3(\epsilon - \varrho) + \kappa\Psi_4 + 2\Phi_{21}(\bar{\varrho} - \epsilon) \\
& - 2\mu\Phi_{10} + 2\pi\Phi_{11} - \bar{\kappa}\Phi_{22} - \Phi_{20}(2\bar{\alpha} - 2\beta - \bar{\pi}) \\
& - 2\bar{\delta}\Lambda, \qquad (12b) \\
\Delta\Psi_1 - \delta\Psi_2 + \bar{\delta}\Phi_{02} - \Delta\Phi_{01} = & \nu\Psi_0 + 2\Psi_1(\gamma - \mu) - 3\tau\Psi_2 + 2\sigma\Psi_3 + 2\Phi_{01}(\bar{\mu} - \gamma) \\
& - 2\varrho\Phi_{12} - \bar{\nu}\Phi_{00} + 2\tau\Phi_{11} + \Phi_{02}\left(\bar{\tau} - 2\bar{\beta} + 2\alpha\right) \\
& + 2\delta\Lambda, \qquad (12c) \\
\bar{\delta}\Psi_0 - D\Psi_1 - \delta\Phi_{00} + D\Phi_{01} = & \Psi_0(4\alpha - \pi) - 2\Psi_1(2\varrho + \epsilon) + 3\kappa\Psi_2 + 2\Phi_{01}(\epsilon + \bar{\varrho}) \\
& + 2\sigma\Phi_{10} - 2\kappa\Phi_{11} - \bar{\kappa}\Phi_{02}, \qquad (12d)
\end{aligned}
$$

Bianchi Identities II

$$\begin{aligned}
\Delta\Psi_2 - \delta\Psi_3 + D\Phi_{22} - \delta\Phi_{21} &= 2\nu\Psi_1 - 3\mu\Psi_2 + 2\Psi_3\left(\beta - \tau\right) + \sigma\Psi_4 + 2\Phi_{21}\left(\bar{\pi} + \beta\right) \\
&\quad - 2\mu\Phi_{11} - \bar{\lambda}\Phi_{20} + 2\pi\Phi_{12} + \Phi_{22}\left(\bar{\varrho} - 2\epsilon - 2\bar{\epsilon}\right) \\
&\quad - 2\Delta\Lambda, \qquad \text{(12e)} \\
D\Psi_4 - \bar{\delta}\Psi_3 - \bar{\delta}\Phi_{21} + \Delta\Phi_{20} &= -3\lambda\Psi_2 + 2\Psi_3\left(2\pi + \alpha\right) - \Psi_4\left(4\epsilon - \varrho\right) + 2\Phi_{21}\left(\alpha - \bar{\tau}\right) \\
&\quad + 2\nu\Phi_{10} + \bar{\sigma}\Phi_{22} - 2\lambda\Phi_{11} - \Phi_{20}\left(\bar{\mu} + 2\gamma - 2\bar{\gamma}\right), \qquad \text{(12f)} \\
\Delta\Psi_0 - \delta\Psi_1 - \delta\Phi_{01} + D\Phi_{02} &= \Psi_0\left(4\gamma - \mu\right) - 2\Psi_1\left(2\tau + \beta\right) + 3\sigma\Psi_2 + 2\Phi_{01}\left(\bar{\pi} - \beta\right) \\
&\quad - 2\kappa\Phi_{12} - \bar{\lambda}\Phi_{00} + 2\sigma\Phi_{11} + \Phi_{02}\left(\bar{\varrho} + 2\epsilon - 2\bar{\epsilon}\right), \qquad \text{(12g)} \\
\Delta\Psi_3 - \delta\Psi_4 - \Delta\Phi_{21} + \bar{\delta}\Phi_{22} &= 3\nu\Psi_2 - 2\Psi_3\left(\gamma + 2\mu\right) - \Psi_4\left(\tau - 4\beta\right) + 2\Phi_{21}\left(\bar{\mu} + \gamma\right) \\
&\quad - 2\nu\Phi_{11} - \bar{\nu}\Phi_{20} + 2\lambda\Phi_{12} + \Phi_{22}\left(\bar{\tau} - 2\alpha - 2\bar{\beta}\right), \qquad \text{(12h)}
\end{aligned}$$

Maxwell's Equations

Field Projections,

$$\phi_0 = F_{\mu\nu}\ell^\mu m^\nu, \quad \phi_1 = \frac{1}{2}F_{\mu\nu}\left(\ell^\mu n^\nu + \bar{m}^\mu m^\nu\right), \quad \phi_2 = F_{\mu\nu}\bar{m}^\mu n^\nu, \tag{13}$$

Field Equations,

$$D\phi_1 - \bar{\delta}\phi_0 = \phi_0\left(\pi - 2\alpha\right) + 2\varrho\phi_1 - \kappa\phi_2 + 2\pi J_\ell, \tag{14a}$$

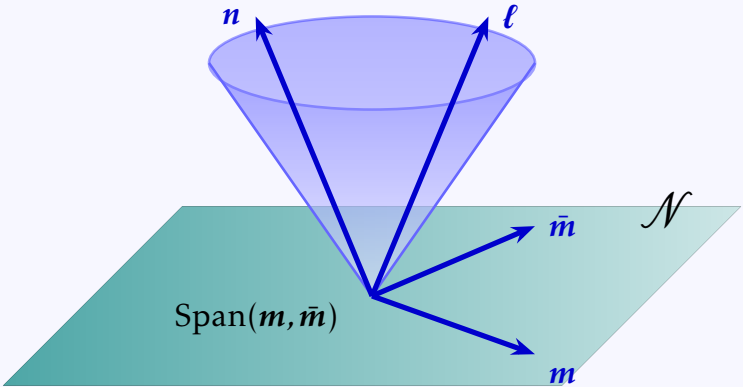
$$\delta\phi_2 - \Delta\phi_1 = \phi_2\left(\tau - 2\beta\right) - \nu\phi_0 + 2\mu\phi_1 + 2\pi J_n, \tag{14b}$$

$$\delta\phi_1 - \Delta\phi_0 = \phi_0\left(\mu - 2\gamma\right) + 2\tau\phi_1 - \sigma\phi_2 + 2\pi J_m, \tag{14c}$$

$$D\phi_2 - \bar{\delta}\phi_1 = \phi_2\left(\varrho - 2\epsilon\right) - \lambda\phi_0 + 2\pi\phi_1 + 2\pi J_{\bar{m}}. \tag{14d}$$

Field Tensor, $\Phi_{\mu\nu} = 2\phi_\mu\bar{\phi}_\nu$

Geometric Picture



Physical Quantities

1. Induced metric, $h_{\mu\nu} = m_\mu \bar{m}_\nu + \bar{m}_\mu m_\nu = 2m_{(\mu}\bar{m}_{\nu)}$.
2. Expansion of $\boldsymbol{\ell}$: $-2\Re\varrho$, $\Im\varrho$ is related to the twist scalar, and σ is the shear of $\boldsymbol{\ell}$.
3. For $\boldsymbol{n}$, the real and imaginary parts of μ give the expansion and twist of $\boldsymbol{n}$, while λ yields its shear.

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Trapped Surfaces

Future trapped surface

Compact cross-section $\mathscr{N}$, containing two systems of null geodesics, $\boldsymbol{\ell}$, $\boldsymbol{n}$, emerging orthogonally from $\mathscr{N}$, towards the future which converge locally, such that they have negative expansions

$$\theta_{(\boldsymbol{\ell})} < 0, \quad \theta_{(\boldsymbol{n})} < 0. \tag{15}$$

Trapped Surfaces

Marginally Outer Trapped Surface (MOTS)

Compact cross-section $\mathscr{N}$, where the expansion of the future-directed, outward pointing normal $\boldsymbol{\ell}$, has vanishing expansion $\theta_{(\boldsymbol{\ell})} = 0$, without restriction on $\boldsymbol{n}$. A surface with $\theta_{(\boldsymbol{\ell})} < 0$ and no restriction on $\boldsymbol{n}$, is called *outer trapped*.

Quasi-Equilibrium cases

Non-Expanding Horizons (NEH)

Null hypersurface $\mathscr{H}$, with topology $\mathscr{H} \simeq \mathbb{R} \times \mathscr{N}$, where $\mathscr{N}$ is a closed submanifold, such that the expansion of $\mathscr{H}$, along any null normal $\boldsymbol{\ell}$ vanishes identically.

All cross sections $\mathscr{N}$ of $\mathscr{H}$ have to be MOTS. Additionally, null energy condition, $T(\boldsymbol{\ell}, \boldsymbol{\ell}) \geq 0$, for any future-directed null normal $\boldsymbol{\ell}$.

Null Normal Equivalence

- $(\boldsymbol{h}, \hat{\boldsymbol{\nabla}})$ characterizes the geometry of $\mathscr{H}$.
- NEH notion is independent of the rescaling of the null normal $\boldsymbol{\ell}$.
  However, rotation 1-form $\boldsymbol{\omega}'_{\mathscr{H}} = \boldsymbol{\omega}_{\mathscr{H}} + \boldsymbol{\Pi}^* d \ln \alpha$, on scaling $\boldsymbol{\ell}' = \alpha \boldsymbol{\ell}$.
- ${}^{\mathscr{H}}\mathcal{L}_{\boldsymbol{\ell}} \boldsymbol{\omega}_{\mathscr{H}} = 0$ depends on the choice of the scaling of $\boldsymbol{\ell}$.

Equivalence Class $[\boldsymbol{\ell}]$

Equivalence class of null normals $[\boldsymbol{\ell}]$, such that $\boldsymbol{\ell}$ and $\tilde{\boldsymbol{\ell}}$ belong to the same class, if $\tilde{\boldsymbol{\ell}} = c\boldsymbol{\ell}$ for some positive constant c, thus helping us chose an explicit choice of the null normal.

Weakly Isolated Horizons

Weakly Isolated Horizon (WIH)

$(\mathscr{H}, [\boldsymbol{\ell}])$ comprises a *weakly isolated horizon (WIH)*, if $\mathscr{H}$ is a NEH, and for each null normal $\boldsymbol{\ell}$ in the equivalence class $[\boldsymbol{\ell}]$, we have

$$ {}^{\mathscr{H}}\mathcal{L}_{\boldsymbol{\ell}}\boldsymbol{\omega}_{\mathscr{H}} = 0, \tag{16} $$

The Lie derivative on the hypersurface,

$$ \begin{aligned} {}^{\mathscr{H}}\mathcal{L}_{\boldsymbol{\ell}}\boldsymbol{\omega}_{\mathscr{H}} &= \boldsymbol{\ell} \cdot \mathrm{d}\boldsymbol{\omega}_{\mathscr{H}} + \mathrm{d}\boldsymbol{\omega}_{\mathscr{H}}(\boldsymbol{\ell})\big|_{\mathscr{H}} \\ &= \boldsymbol{\ell} \cdot \mathrm{d}\boldsymbol{\omega}_{\mathscr{H}} + \hat{\nabla}\kappa_{(\boldsymbol{\ell})} = 0. \end{aligned} \tag{17} $$

Motivating Isolated Horizons

Intrinsic nature of the affine connection, $\hat{\nabla}\boldsymbol{\ell} = \boldsymbol{\ell} \otimes \boldsymbol{\omega}_{\mathscr{H}}$, such that,

$$\begin{aligned} {}^{\mathscr{H}}\mathcal{L}_{\boldsymbol{\ell}}\left(\hat{\nabla}\boldsymbol{\ell}\right) &= {}^{\mathscr{H}}\mathcal{L}_{\boldsymbol{\ell}}\boldsymbol{\ell} \otimes \boldsymbol{\omega}_{\mathscr{H}} + \boldsymbol{\ell} \otimes {}^{\mathscr{H}}\mathcal{L}_{\boldsymbol{\ell}}\boldsymbol{\omega}_{\mathscr{H}} \\ &= \left({}^{\mathscr{H}}\mathcal{L}_{\boldsymbol{\ell}}\boldsymbol{\ell}\right) \otimes \boldsymbol{\omega}_{\mathscr{H}} = \hat{\nabla}\left({}^{\mathscr{H}}\mathcal{L}_{\boldsymbol{\ell}}\boldsymbol{\ell}\right), \end{aligned} \tag{18a}$$

hence we have retrieve the identity,

$$\left[{}^{\mathscr{H}}\mathcal{L}_{\boldsymbol{\ell}}, \hat{\nabla}\right]\boldsymbol{\ell} = 0. \tag{18b}$$

Restriction on the intrinsic nature of the horizon, described by the induced affine connection, $\hat{\nabla}$.

Isolated Horizons

Isolated Horizon

$(\mathscr{H}, [\boldsymbol{\ell}])$ constitutes an *Isolated Horizon (IH)*, given $\mathscr{H}$ is a NEH, and each null normal $\boldsymbol{\ell}$ in $[\boldsymbol{\ell}]$ satisfies,

$$\left[{}^{\mathscr{H}}\mathcal{L}_{\boldsymbol{\ell}}, \hat{\nabla}\right] = 0. \tag{19}$$

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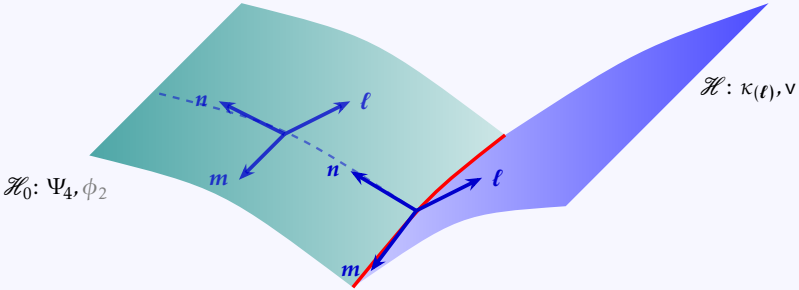
Characteristic Initial Value



Frame Functions

Near-Horizon extension of local null tetrad,

$$\ell^i \nabla_i = D = \partial_{\mathsf{v}} + \mathsf{U}\partial_{\mathsf{r}} + \mathsf{X}^i \partial_{\mathsf{x}^i}, \tag{20a}$$

$$n^i \nabla_i = \Delta = -\partial_{\mathsf{r}}, \tag{20b}$$

$$m^i \nabla_i = \delta = \Omega \partial_{\mathsf{r}} + \xi^i \partial_{\mathsf{x}^i}. \tag{20c}$$

Horizon conditions: $\mathsf{U} \doteq 0$, $\mathsf{X}^i \doteq 0$, $\Omega \doteq 0$.

Spin Coefficients

$\boldsymbol{n}$ affinely paramterized, and $\boldsymbol{\ell}$, $\boldsymbol{m}$ are parallel propagated along $\boldsymbol{n}$,

$$\tau = \nu = \gamma = 0. \tag{21}$$

Commutation Relations,

$$\mu = \bar{\mu}, \quad \pi = \alpha + \bar{\beta}. \tag{22}$$

Geodesic, hypersurface-orthogonal and expansion-free nature of $\boldsymbol{\ell}$,

$$\kappa \doteq \varrho \doteq 0, \tag{23}$$

Raychaudhuri's Equation constrains the shear-free nature,

$$\sigma \doteq 0. \tag{24}$$

Near Horizon Field Equations

Time evolution equations containing the directional derivative D,

$$D\varrho - \bar{\delta}\kappa = \varrho^2 + |\sigma|^2 + \varrho\left(\epsilon + \bar{\epsilon}\right) - 2\kappa\alpha, \tag{25a}$$

$$D\sigma - \delta\kappa = \sigma\left(\epsilon + \bar{\epsilon} + \varrho + \bar{\varrho}\right) - 2\kappa\beta + \Psi_0, \tag{25b}$$

$$D\alpha - \bar{\delta}\epsilon = \alpha\left(\varrho + \bar{\epsilon} - 2\epsilon\right) + \beta\bar{\sigma} - \bar{\beta}\epsilon - \kappa\lambda + \pi\left(\epsilon + \varrho\right), \tag{25c}$$

$$D\beta - \delta\epsilon = \sigma\left(\alpha + \pi\right) + \beta\left(\bar{\varrho} - \bar{\epsilon}\right) - \kappa\mu + \epsilon\left(\bar{\pi} - \bar{\alpha}\right) + \Psi_1, \tag{25d}$$

$$D\lambda - \bar{\delta}\pi = \lambda\left(\varrho - 2\epsilon\right) + \bar{\sigma}\mu + 2\pi\alpha, \tag{25e}$$

$$D\mu - \delta\pi = \mu\left(\bar{\varrho} - \epsilon - \bar{\epsilon}\right) + \sigma\lambda + 2\pi\beta + \Psi_2, \tag{25f}$$

Angular Field Equations

$$\delta\alpha - \bar{\delta}\beta = \mu\varrho - \lambda\sigma + |\alpha|^2 + |\beta|^2 - 2\alpha\beta - \Psi_2, \tag{26a}$$

$$\delta\varrho - \bar{\delta}\sigma = \varrho\bar{\pi} + \sigma\left(\bar{\beta} - 3\alpha\right) - \Psi_1, \tag{26b}$$

$$\delta\lambda - \bar{\delta}\mu = \mu\pi + \lambda\left(\bar{\alpha} - 3\beta\right) - \Psi_3. \tag{26c}$$

Radial Field Equations *(Decoupled)*

$$\Delta\pi = -\mu\pi - \lambda\bar{\pi} - \Psi_3, \tag{27a}$$
$$\Delta\beta = -\beta\mu - \alpha\bar{\lambda}, \tag{27b}$$
$$\Delta\alpha = -\lambda\beta - \alpha\mu - \Psi_3, \tag{27c}$$
$$\Delta\lambda = -2\lambda\mu - \Psi_4, \tag{27d}$$
$$\Delta\mu = -\mu^2 - |\lambda|^2, \tag{27e}$$
$$\Delta\varrho = -\varrho\mu - \sigma\lambda - \Psi_2, \tag{27f}$$
$$\Delta\sigma = -\mu\sigma - \bar{\lambda}\varrho, \tag{27g}$$
$$\Delta\kappa = -\varrho\bar{\pi} - \sigma\pi - \Psi_1, \tag{27h}$$
$$\Delta\epsilon = -\alpha\bar{\pi} - \beta\pi - \Psi_2, \tag{27i}$$

Near Horizon Bianchi Identities

Evolution equations,

$$D\Psi_1 - \bar{\delta}\Psi_0 = \Psi_0\left(\pi - 4\alpha\right) + 2\Psi_1\left(2\varrho + \epsilon\right) - 3\kappa\Psi_2, \tag{28a}$$

$$D\Psi_2 - \bar{\delta}\Psi_1 = -\lambda\Psi_0 - 2\Psi_1\left(\pi - \alpha\right) + 3\varrho\Psi_2 - 2\kappa\Psi_3, \tag{28b}$$

$$D\Psi_3 - \bar{\delta}\Psi_2 = -2\lambda\Psi_1 + 3\pi\Psi_2 - 2\Psi_3\left(\epsilon - \varrho\right) - \kappa\Psi_4, \tag{28c}$$

$$D\Psi_4 - \bar{\delta}\Psi_3 = -3\lambda\Psi_2 + 2\Psi_3\left(2\pi + \alpha\right) - \Psi_4\left(4\epsilon - \varrho\right), \tag{28d}$$

Radial Bianchi Identities

$$\Delta\Psi_0 - \delta\Psi_1 = -\mu\Psi_0 - 2\beta\Psi_1 + 3\sigma\Psi_2, \tag{29a}$$
$$\Delta\Psi_1 - \delta\Psi_2 = -2\mu\Psi_1 + 2\sigma\Psi_3, \tag{29b}$$
$$\Delta\Psi_2 - \delta\Psi_3 = -3\mu\Psi_2 + 2\beta\Psi_3 + \sigma\Psi_4, \tag{29c}$$
$$\Delta\Psi_3 - \delta\Psi_4 = -4\mu\Psi_3 + 4\beta\Psi_4. \tag{29d}$$

Since no radial derivative component for Ψ_4, given as free data that we provide on the null hypersurface $\mathscr{H}_0$.

Schematic

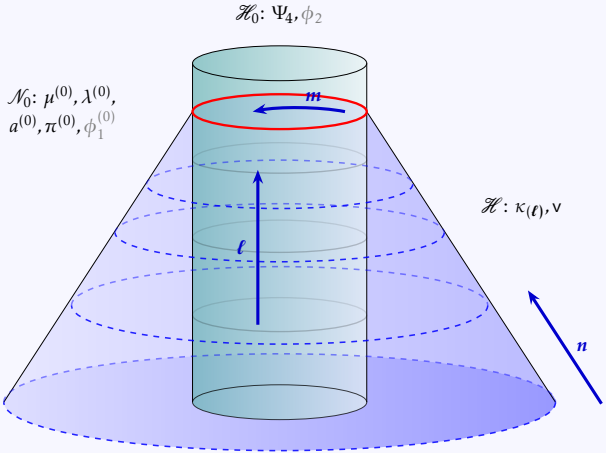
Hamiltonian Calculations
Mass M, Surface gravity κ

Initial Data, Gauge Constraints
NEH/WIH, Rotation 1-form, Geodesic Nature

Multipole Moments, Algebraic Weyl properties
Intrinsic Geometry Characterization
Constraints on the Horizon

Newman-Penrose Field equations
Evolution & Constraint Equations
Radial Series Expansion

Spherical Background



Schwarzschild Background

Hamiltonian Calculations

$\mathsf{M} = \frac{1}{2}\mathsf{R}$, $\kappa_{(\boldsymbol{\ell})} = \frac{1}{2\mathsf{R}}$

Initial Data, Gauge Constraints

$\tau = \nu = \gamma = 0$, $\mu = \bar{\mu}$, $\pi = \alpha + \bar{\beta}$

$\kappa^{(0)} = \varrho^{(0)} = \sigma^{(0)} = 0$, Gauge choice: $\epsilon = \bar{\epsilon}$

Initial Data: $\Psi_4^{(0)} = 0$, $D\mu^{(0)} = 0$, $\lambda^{(0)} = 0$

Multipole Moments, Algebraic Weyl properties

Petrov Type D Horizon: $\Psi_0^{(0)} = \Psi_1^{(0)} = \Psi_3^{(0)} = \Psi_4^{(0)} = 0$

${}^{(2)}R = -4\Re\Psi_2^{(0)}$, $\Psi_2^{(0)} = -\frac{1}{8\mathsf{M}^2}$

Newman-Penrose Field equations

Evolution & Constraint Equations

Radial Series Expansion

Schwarzschild Solution

$$\tau = \nu = \lambda = \sigma = \pi = \kappa = 0, \tag{30a}$$

$$\mu = -\frac{1}{\mathsf{r}}, \quad \epsilon = \frac{\mathsf{R}}{4\mathsf{r}^2}, \quad \varrho = -\frac{1}{2\mathsf{r}}\left(1 - \frac{\mathsf{R}}{\mathsf{r}}\right), \tag{30b}$$

$$a = \frac{a^{(0)}}{\mathsf{r}} = \frac{\bar{\xi}}{\sqrt{2}\mathsf{R}\mathsf{r}} \tag{30c}$$

$$D = \partial_{\mathsf{v}} - \left(1 - \frac{\mathsf{R}}{\mathsf{r}}\right)\partial_{\mathsf{r}}, \quad \Delta = -\partial_{\mathsf{r}}, \quad \delta = \frac{P}{\mathsf{r}}\partial_{\xi} \tag{31}$$

$$\Psi_0^{[0]} = \Psi_1^{[0]} = \Psi_3^{[0]} = \Psi_4^{[0]} = 0, \quad \Psi_2^{[0]} = -\frac{\mathsf{R}}{2\mathsf{r}^3}. \tag{32}$$

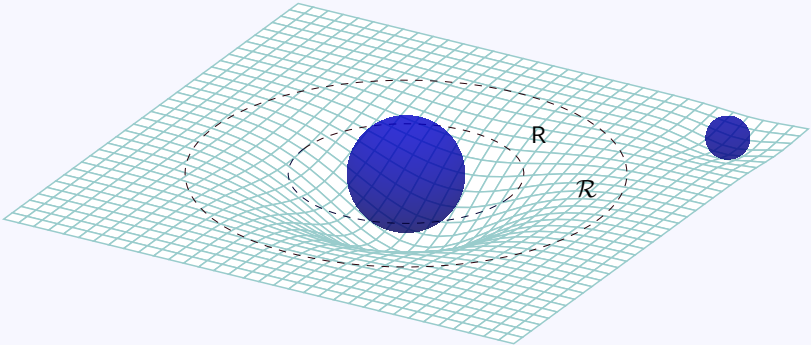
Electromagnetism

- Charge Multipoles
- Raychaudhuri's Equation, $\phi_0^{(0)} = 0$.
- Maxwell's equations, $D\phi_1^{(0)} = 0$, Free Data: $\phi_2^{(0)}$.
- $\Re\Psi_2^{(0)} \to \Re\Psi_2^{(0)} + 2\left|\phi_1^{(0)}\right|^2$, $\Psi_3^{(0)} \to \Psi_3^{(0)} + 2\phi_1^{(0)}\bar{\phi}_2^{(0)}$.

Outline

1. Preliminaries
2. Isolated Horizons
3. Probing Near-Horizon Structure
4. Tidal Deformabilities

Tidally Perturbed Scenario



Newtonian Theory of Deformabilities

- Spherical Newtonian body of mass M and equilibrium radius R.
- Embedded in an external gravitational potential, $\Phi_{\rm ext}(\mathsf{t}, \vec{\mathsf{x}})$.
- Symmetric, and trace-free tidal moments,

$$\mathcal{E}_L(\mathsf{t}) = -\frac{1}{(\ell-2)!}\partial_{\langle L\rangle}\Phi_{\rm ext}(\mathsf{t}, \vec{\mathsf{x}}_{\mathsf{COM}}). \tag{33}$$

for $0 \leq \mathsf{r} \leq \mathcal{R}$.

Deformed Gravitational Potential

Multipole moments of the deformed body, for $r \geq R$,

$$I_L(t) = \int_{\mathbb{R}^3} d^3x \; \delta\varrho(t, \vec{x}) x^{\langle L \rangle}, \tag{34}$$

where $\delta\varrho(t, \vec{x})$ is the mass density perturbation induced by the applied tidal field.

Total Gravitational Potential

Response,

$$\Phi(\mathsf{t},\vec{\mathsf{x}}) = -\sum_{\ell=2}^{\infty}\frac{1}{\ell(\ell-1)}\mathsf{x}^L\mathcal{E}_L(\mathsf{t}) + \frac{\mathsf{M}}{\mathsf{r}} + \sum_{\ell=2}^{\infty}\frac{(2\ell-1)!!}{\ell!}\frac{\mathsf{n}^L I_L(t)}{\mathsf{r}^{\ell+1}}, \tag{35}$$

with $\mathsf{r} = |\vec{\mathsf{x}}|$, and $\vec{\mathsf{n}} = \frac{1}{\mathsf{r}}\vec{\mathsf{x}}$.

Love Numbers

- Adiabatic approximation, Tidal deformability parameter,

$$I_L(\mathrm{t}) = -\lambda_\ell \mathcal{E}_L(\mathrm{t}), \tag{36}$$

  scales as $\mathrm{R}^{2\ell+1}$.
- Dimensionless tidal love numbers,

$$\lambda_\ell = \frac{2(\ell-2)!}{(2\ell-1)!!} k_\ell \mathrm{R}^{2\ell+1}, \tag{37}$$

- Potential,

$$\Phi(\mathrm{t}, \mathrm{r}, \xi, \bar{\xi}) = \frac{\mathrm{M}}{\mathrm{r}} - \sum_{\ell,m} \frac{1}{\ell(\ell-1)} \mathrm{r}^\ell \left[1 + 2k_\ell \left(\frac{\mathrm{R}}{\mathrm{r}}\right)^{2\ell+1}\right] \mathcal{E}_{\ell,m}(\mathrm{t}) Y_{\ell,m}(\xi, \bar{\xi}). \tag{38}$$

Relativistic Theory of Deformabilities

$$g_{00} = -1 + \frac{2\Phi}{c^2} + \mathcal{O}(c^{-4}), \tag{39a}$$

$$g_{0\mathrm{i}} = \mathcal{O}(c^{-3}), \tag{39b}$$

$$g_{\mathrm{ij}} = \delta_{\mathrm{ij}} \left(1 + \frac{2\Phi}{c^2}\right) + \mathcal{O}(c^{-4}). \tag{39c}$$

Surficial Love Numbers

- Deformation in Radius,

$$\hat{\mathsf{R}}(\mathsf{t},\xi,\bar{\xi}) = -\mathsf{R}\sum_{\ell,m}\frac{1}{\ell(\ell-1)}h_\ell\mathcal{E}_{\ell,m}(\mathsf{t})Y_{\ell,m}(\xi,\bar{\xi}), \tag{40}$$

  where the surficial Love numbers h_ℓ are dimensionless.

- Curvature,

$$^{(2)}\hat{R}(\mathsf{t},\xi,\bar{\xi}) = -\frac{2}{\mathsf{R}^2}\sum_{\ell,m}\frac{\ell+2}{\ell}h_\ell\mathcal{E}_{\ell,m}(\mathsf{t})Y_{\ell,m}(\xi,\bar{\xi}). \tag{41}$$

Tidal Perturbations of Isolated Horizons

- Generic Cross-Section metric,

$$ds^2 \doteq \frac{2\mathsf{R}^2}{|P|^2} d\xi d\bar{\xi}. \tag{42}$$

- Tidal Perturbation,

$$P \doteq P_{[0]} \left(1 + \hat{P}\right), \tag{43}$$

where $\hat{P}$ is a small perturbation, which we expand up to linear order.

- Perturbed Intrinsic Geometry, Curvature,

$$^{(2)}\hat{R} \doteq -4\Re\hat{\Psi}_2 \doteq 2\Delta_{[0]}\Re\hat{P} + 2\Re\hat{P}^{(2)}R_{[0]}. \tag{44}$$

where $\hat{\Psi}_2 \doteq \sum_{\ell,m} \hat{k}_{\ell,m} Y_{\ell,m}(\xi, \bar{\xi})$.

Spherical Background

- Perturbed horizon, characterized by a surface of vanishing expansion and shear.
- $D_{[0]}\hat{\pi} \doteq D_{[0]}\hat{\alpha} \doteq D_{[0]}\hat{\beta} \doteq D_{[0]}\hat{\Psi}_1 \doteq D_{[0]}\hat{\Psi}_2 \doteq 0.$
- Foliation Independency of $\hat{\Psi}_4$,

$$D^2_{[0]}\hat{\Psi}_4 + \frac{3}{2\mathsf{R}}D_{[0]}\hat{\Psi}_4 + \frac{1}{2\mathsf{R}^2}\hat{\Psi}_4 \doteq \bar{\eth}^2_{[0]}\hat{\Psi}_2. \tag{45}$$

- Teukolsky Equation,

$$\begin{aligned}\mathcal{O}^{\mathsf{T}}_{[0]}\hat{\Psi}_4 = \Big(\Delta D_{[0]} - \bar{\eth}_{[0]}\eth_{[0]} + 5\mu_{[0]}D_{[0]} + 5\mu_{[0]}\left(4\epsilon_{[0]} - \varrho_{[0]}\right) + \Delta\left(4\epsilon_{[0]} - \varrho_{[0]}\right) \\ + \left(4\epsilon_{[0]} - \varrho_{[0]}\right)\Delta - 3\Psi_2^{[0]}\Big)\hat{\Psi}_4 = 0.\end{aligned} \tag{46}$$

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Thank You!

Questions? Comments?

Appendix: Schwarzschild Solution

- Spacetime near the horizon, $\tau = \nu = \gamma = 0$, $\mu = \bar{\mu}$, $\pi = \alpha + \bar{\beta}$.
- Extended gauge choice, $\epsilon = \bar{\epsilon}$, $\kappa^{(0)} = \varrho^{(0)} = \sigma^{(0)} = 0$.

Characteristic Initial Data

- Trivial choice of initial data given by $\Psi_4^{(0)} = 0$, on the transverse null surface, $\mathscr{H}_0$, and choose the common cross section $\mathscr{N}_0$.
- For expansion, $\mu^{(0)}$ and shear, $\lambda^{(0)}$ of $\boldsymbol{n}$, we extend the spherical symmetry to $\mathscr{N}_0$ to have $D\mu^{(0)} = 0$ as an angular function, and $\lambda^{(0)} = 0$.

Petrov Classification

At the Schwarzschild horizon, we assume the metric is Petrov Type D, to arrive at,

$$\Psi_0^{(0)} = \Psi_1^{(0)} = \Psi_3^{(0)} = \Psi_4^{(0)} = 0, \tag{47}$$

on the horizon. Further, $\Psi_2^{(0)}$ is an angular function and assumes spherical symmetry.

Horizon Multipole Moments

- The choice of $\Psi_2^{(0)}$ determines the horizon multipole moments, specifically the mass multipole moments, since we have a non-spinning solution.
- $\Psi_2^{(0)}$, which is constant, must be completely real, since the angular momentum multipoles, related to $\Im\Psi_2^{(0)}$, vanish.
- Hamiltonian Constraints $\mathrm{M} = \frac{1}{2}\mathrm{R}$, and Surface Gravity, $\kappa_{(\boldsymbol{\ell})} = \frac{1}{2\mathrm{R}}$.

Scalar Curvature

Gauss-Bonnet theorem with the Ricci scalar relating Gaussian Curvature,

$$\oint_{\mathscr{N}_0} {}^{(2)}R\, \boldsymbol{\epsilon} = 8\pi, \tag{48a}$$

hence, we use the constancy of $\Psi_2^{(0)}$, and the area of the horizon $\mathsf{A} = 4\pi\mathsf{R}^2$, to arrive at,

$$\Psi_2^{(0)} = \Re\Psi_2^{(0)} = -\frac{1}{2\mathsf{R}^2} = -\frac{1}{8\mathsf{M}^2}, \tag{48b}$$

thus determining the intrinsic horizon geometry through the topology of $\mathscr{N}_0$.

Expansion and Shear

Angular function $\mu^{(0)}$ to $\Psi_2^{(0)}$ by the time evolution field equations,

$$\kappa_{(\boldsymbol{\ell})}\mu^{(0)} = \eth\pi^{(0)} + \left|\pi^{(0)}\right|^2 + \Psi_2^{(0)}, \tag{49}$$

where $\pi^{(0)} = 0$, such that,

$$\mu^{(0)} = -\frac{1}{2\mathrm{M}}. \tag{50}$$

Radial Derivatives

$$\mu^{(1)} = -\frac{d\mu}{d\mathsf{r}} = -\mu^2 - |\lambda|^2, \tag{51a}$$

$$\lambda^{(1)} = -\frac{d\lambda}{d\mathsf{r}} = 2\lambda\mu, \tag{51b}$$

$$\sigma^{(1)} = -\frac{d\sigma}{d\mathsf{r}} = -\mu\sigma - \bar{\lambda}\varrho. \tag{51c}$$

Expansion and Shear

We use $\lambda^{(0)} = 0$, $\mu^{(0)} = -\frac{1}{2\mathsf{M}}$, and $\sigma^{(0)} = 0$,

$$\lambda = 0, \tag{52a}$$

$$\mu = -\frac{1}{\mathsf{r}}, \tag{52b}$$

$$\sigma = 0. \tag{52c}$$

Radial Bianchi Identities

$$\Delta\Psi_3 = \frac{4}{\mathsf{r}}\Psi_3, \tag{53a}$$

$$\Delta\Psi_2 - \delta\Psi_3 = \frac{3}{\mathsf{r}}\Psi_2 + 2\beta\Psi_3, \tag{53b}$$

$$\Delta\Psi_1 - \delta\Psi_2 = \frac{2}{\mathsf{r}}\Psi_1, \tag{53c}$$

$$\Delta\Psi_0 - \delta\Psi_1 = \frac{1}{\mathsf{r}}\Psi_0 - 2\beta\Psi_1. \tag{53d}$$

Weyl Tensors

We have $\Psi_3^{(0)} = 0$, and $\Psi_2^{(0)} = -\frac{1}{2R^2}$, to arrive at

$$\Psi_3 = 0, \tag{54a}$$

$$\Psi_2 = -\frac{M}{r^2}, \tag{54b}$$

$$\Psi_1 = 0, \tag{54c}$$

$$\Psi_0 = 0. \tag{54d}$$

Radial Derivatives

$$\pi^{(1)} = -\frac{d\pi}{d\mathsf{r}} = \frac{1}{\mathsf{r}}\pi, \tag{55a}$$

$$\kappa^{(1)} = -\frac{d\kappa}{d\mathsf{r}} = -\varrho\bar{\pi}, \tag{55b}$$

$$\varrho^{(1)} = -\frac{d\varrho}{d\mathsf{r}} = \frac{1}{\mathsf{r}}\varrho + \frac{\mathsf{M}}{\mathsf{r}^3}, \tag{55c}$$

$$\epsilon^{(1)} = -\frac{d\epsilon}{d\mathsf{r}} = -\alpha\bar{\pi} - \beta\pi + \frac{\mathsf{M}}{\mathsf{r}^3}, \tag{55d}$$

$$\alpha^{(1)} - \bar{\beta}^{(1)} = -\frac{d\left(\alpha - \bar{\beta}\right)}{d\mathsf{r}} = \frac{1}{\mathsf{r}}\left(\alpha - \bar{\beta}\right), \tag{55e}$$

Spin Coefficients

Horizon Constraints: $\pi^{(0)} = 0$, $\epsilon^{(0)} = \frac{1}{2}\kappa_{(\boldsymbol{\ell})} = \frac{1}{8\mathsf{M}}$, $\varrho^{(0)} = 0$,

$$\pi = 0, \tag{56a}$$

$$\kappa = 0, \tag{56b}$$

$$\varrho = -\frac{1}{2\mathsf{r}}\left(1 - \frac{2\mathsf{M}}{\mathsf{r}}\right), \tag{56c}$$

$$\epsilon = \frac{\mathsf{M}}{2\mathsf{r}^2}, \tag{56d}$$

$$\alpha - \bar{\beta} = \frac{\alpha^{(0)} - \bar{\beta}^{(0)}}{\mathsf{r}}. \tag{56e}$$

Frame Functions

$$\Delta \mathsf{U} = -\frac{d\mathsf{U}}{d\mathsf{r}} = \frac{\mathsf{M}}{\mathsf{r}^2}, \tag{57a}$$

$$\Delta \Omega = -\frac{d\Omega}{d\mathsf{r}} = -\frac{1}{\mathsf{r}}\Omega, \tag{57b}$$

$$\Delta \xi^i = -\frac{d\xi^i}{d\mathsf{r}} = -\frac{1}{\mathsf{r}}\xi^i, \tag{57c}$$

$$\Delta \mathsf{X}^i = -\frac{d\mathsf{X}^i}{d\mathsf{r}} = 0. \tag{57d}$$

Frame Functions

Horizon: $\mathsf{U}^{(0)} = \mathsf{X}^{(0)} = \Omega^{(0)} = 0$, $\xi^{(0),i}$ is determined by the holomorphic coordinates,

$$\mathsf{U} = \frac{2\mathsf{M} - \mathsf{r}}{2\mathsf{r}}, \tag{58a}$$

$$\Omega = 0, \tag{58b}$$

$$\xi^i = \xi^{(0),i}\mathsf{r}, \tag{58c}$$

$$\mathsf{X}^i = 0. \tag{58d}$$

Null Tetrad

Null Tetrad,

$$\ell^i \nabla_i = D = \partial_{\mathsf{v}} - \left(1 - \frac{2\mathsf{M}}{\mathsf{r}}\right)\partial_{\mathsf{r}}, \tag{59a}$$

$$n^i \nabla_i = \Delta = -\partial_{\mathsf{r}}, \tag{59b}$$

$$m^i \nabla_i = \delta = \mathsf{r}\xi^{(0),i}\partial_{\mathsf{x}^i}, \tag{59c}$$

Contravariant Metric,

$$g^{\mathsf{rr}} = 2\mathsf{U} = \frac{2\mathsf{M}}{\mathsf{r}} - 1, \tag{60a}$$

$$g^{\mathsf{vr}} = 1, \tag{60b}$$

$$g^{\mathsf{r}\mathsf{x}^i} = 0, \tag{60c}$$

$$g^{\mathsf{x}^i\mathsf{x}^j} = \xi^i\bar{\xi}^j + \bar{\xi}^i\xi^j = 2\Re\{\xi^{(0),i}\bar{\xi}^{(0),j}\}\mathsf{r}^2. \tag{60d}$$

Spacetime Metric

Null Co-tetrad,

$$\boldsymbol{\ell} = -\frac{1}{2}\left(1 - \frac{2\mathsf{M}}{\mathsf{r}}\right) d\mathsf{v} + d\mathsf{r}, \tag{61a}$$

$$\boldsymbol{n} = -d\mathsf{v}, \tag{61b}$$

$$\boldsymbol{m} = \frac{1}{2}\mathsf{r}\xi_i^{(0)} d\mathsf{x}^i, \tag{61c}$$

Covariant Metric, $g_{\mu\nu} = 2\left(-\ell_{(\mu}n_{\nu)} + m_{(\mu}\bar{m}_{\nu)}\right)$,

$$ds^2 = -\left(1 - \frac{2\mathsf{M}}{\mathsf{r}}\right) d\mathsf{v}^2 + 2d\mathsf{v}d\mathsf{r} + \frac{1}{2\mathsf{r}^2}\Re\{\xi_i^{(0)}\bar{\xi}_j^{(0)}\} d\mathsf{x}^i d\mathsf{x}^j \tag{62}$$